

High-Level Requirements – Compiled List

1. Functional Requirements

Functional requirements describe **what the system must do**.

Sales Management

1. The system shall allow a cashier to **process sales transactions** for electronic products and gadgets.
2. The system shall **calculate the total price automatically** based on selected items and quantities.
3. The system shall **generate a digital or printed receipt** after each completed sale.
4. The system shall allow the cashier to **cancel or modify a transaction before final confirmation**.
5. The system shall **record each completed sale** in the sales database.

Inventory Management

6. The system shall allow staff to **add new electronic products or gadgets to the inventory**.
7. The system shall allow staff to **update product details** such as price, description, and quantity.
8. The system shall allow staff to **remove discontinued products** from the inventory.
9. The system shall **automatically reduce inventory quantity** when a product is sold.
10. The system shall **notify staff when stock levels are low**.

User Management

11. The system shall allow administrators to **create user accounts** for employees.
12. The system shall allow users to **log in using a username and password**.
13. The system shall support **different user roles** such as cashier and administrator.
14. The system shall allow administrators to **disable or remove user accounts**.

Reporting

15. The system shall allow administrators to **view daily sales reports**.

16. The system shall allow administrators to **generate reports for weekly or monthly sales.**
17. The system shall allow administrators to **view inventory reports** showing current stock levels.
18. The system shall allow administrators to **export reports for business analysis.**

Product Search

19. The system shall allow users to **search for products by name or category.**
20. The system shall allow users to **view product details including price and stock availability.**

2. Non-Functional Requirements

Non-functional requirements describe **how well the system performs its functions.**

Performance

1. The system should process a sales transaction **within 2 seconds.**
2. The system should support **multiple transactions without performance degradation.**

Security

3. The system shall require **user authentication before access.**
4. Sensitive data such as passwords shall be **securely stored using encryption or hashing.**
5. Only authorized users shall be able to **access administrative functions.**

Usability

6. The system interface should be **simple and easy for cashiers to use.**
7. The system should require **minimal training for new employees.**

Reliability

8. The system should ensure **data accuracy for sales and inventory records.**

9. The system should maintain **consistent data even during system interruptions**.

Scalability

10. The system should support **future expansion to multiple store branches**.

Maintainability

11. The system should be **easy to update or modify when new features are required**.

Availability

12. The system should be **available during business operating hours with minimal downtime**.

Use Cases – Compiled List

Use Case Name	Primary Actor	Short Description
Login to System	Cashier / Administrator	The user enters login credentials to gain access to the POS system.
Process Sale	Cashier	The cashier selects or scans items and completes a sales transaction for a customer.
Generate Receipt	Cashier	The system generates and displays or prints a receipt after a transaction is completed.
Cancel Transaction	Cashier	The cashier cancels an ongoing transaction before it is finalized.
Modify Transaction	Cashier	The cashier adds or removes items from a transaction before completing the sale.
Search Product	Cashier	The cashier searches for products by name, category, or product ID.
View Product Details	Cashier	The system displays information about a product including price and stock availability.

Use Case Name	Primary Actor	Short Description
Add Product	Administrator / Inventory Staff	Authorized staff add new electronic products or gadgets to the inventory.
Update Product Information	Administrator / Inventory Staff	Authorized staff update product details such as price, description, or stock quantity.
Remove Product	Administrator	The administrator removes products that are discontinued or no longer available.
Manage User Accounts	Administrator	The administrator creates, updates, or disables user accounts in the system.
View Sales Report	Administrator	The administrator generates and views reports of sales transactions over a period of time.
View Inventory Report	Administrator / Inventory Staff	The system displays the current inventory levels of products in the store.
Receive Low Stock Alert	System	The system automatically alerts staff when product stock levels fall below a defined threshold.
Export Sales Data	Administrator	The administrator exports sales data for business analysis or record keeping.

***Compare the above use cases with the 8 in the slides